NATURAL RESOURCES

Dr. Prasenjit Adak

Resources

- What are resources?
  - A resource is a source or supply from which a benefit is produced and that has some utility.
  - Example: Land, Minerals Soil, Water, Energy, Food etc.
- Types of resources:
  - Natural resources
  - Man-made resources
  - Human resources

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Natural Resources

- What are natural resources?
  - The natural resources may be defined as any material given to us by nature which can be transformed in a way that it becomes more valuable and useful.
- Types of natural resources:
  - Non-renewable resources
  - Renewable resources

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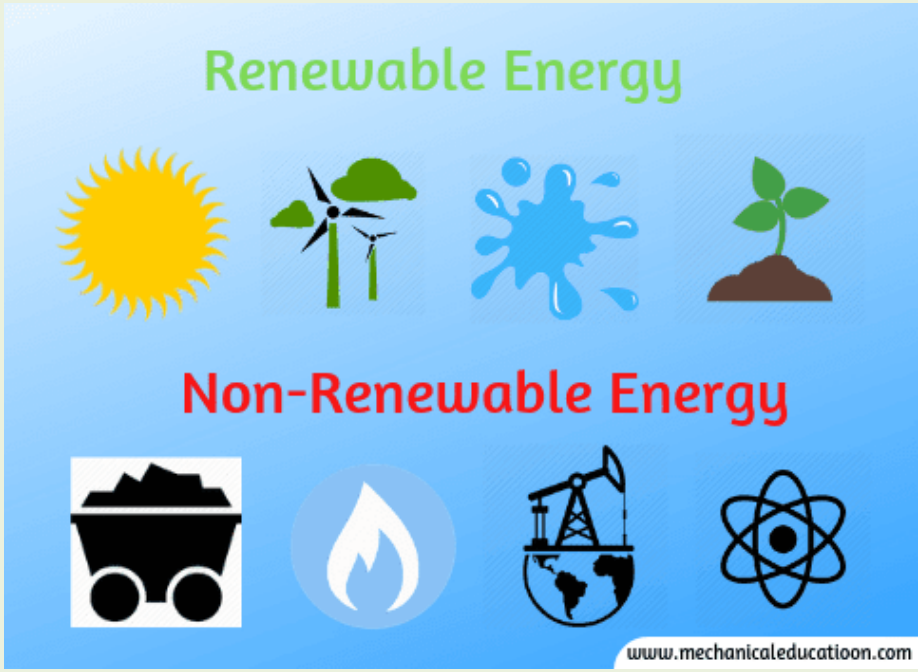
Renewable and Non-renewable Resources

- What are renewable resources?
  - The resources which are renewed or replenished fast or have unlimited source.
  - Example: Water, air, biomass, solar energy etc.
- What are non-renewable resources?
  - The resources which can not be renewed or replenished.
  - Example: Fossil fuels, Nuclear power, Minerals etc.



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Some natural resources to be discussed

Land Resource

Forest Resource

Water Resource

Energy Resource

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LAND RESOURCE

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Land as a resource

- The study of soils in their natural environment is called “pedology”.
- Land is a renewable but limited resource
- Problems related to land resource
  - Reaching the carrying capacity
  - Population density
  - Improper land-use planning
  - Saltwater intrusion
  - Land degradation
  - Soil erosion

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Land Degradation

- Causes
  - Natural causes
    - Heavy rainfall
    - High speed winds
    - Natural disasters: earthquake, landslide, flood, draught
    - Expansion of desert
  - Anthropogenic causes
    - Mining
    - Urbanization
    - Deforestation
    - Overgrazing
    - Water logging
    - Construction of dams
    - Extensive use of fertilizers
    - Dumping of industrial and municipal wastes

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Soil

- Soil is a dynamic natural body capable of supporting a vegetative cover. It is composed largely of weathered rocks, water, oxygen and organic materials.
- Soil formation
  - Processes
    - Physical weathering
    - Chemical weathering
    - Biological weathering
  - Factors
    - Parent material
    - Living organisms
    - Climate
    - Topography
    - Time

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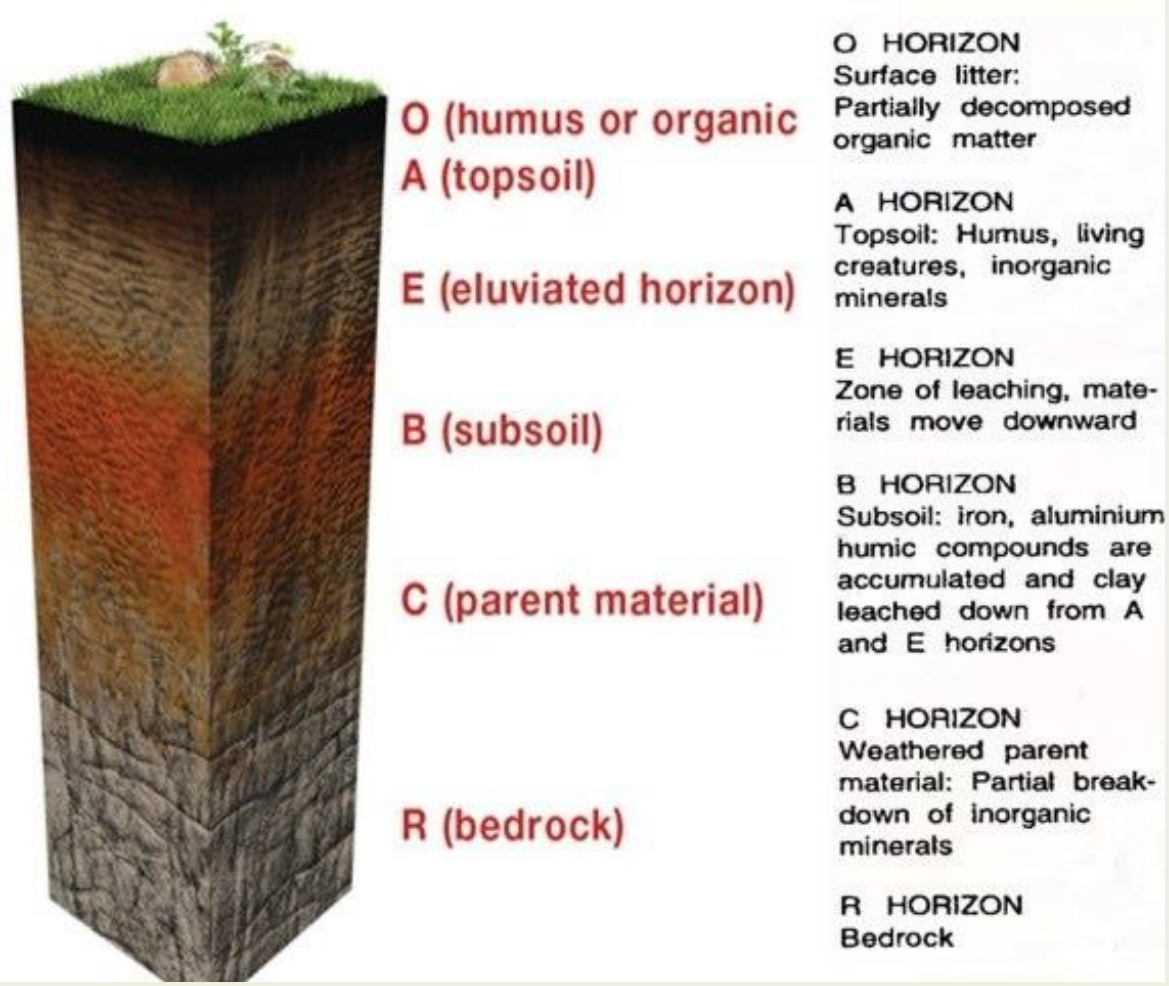
Soil

- Soil profile



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Soil

- Functions of soil
  - Facilitates nutrient cycle
  - Food and other biomass production
  - Stores water and regulates water supply
  - Regulates the emission of trace gases
  - Filter ground water
  - Degrades pollutants
  - Biological habitat
  - Source of clay
  - Platform for man-made structures: buildings, highways

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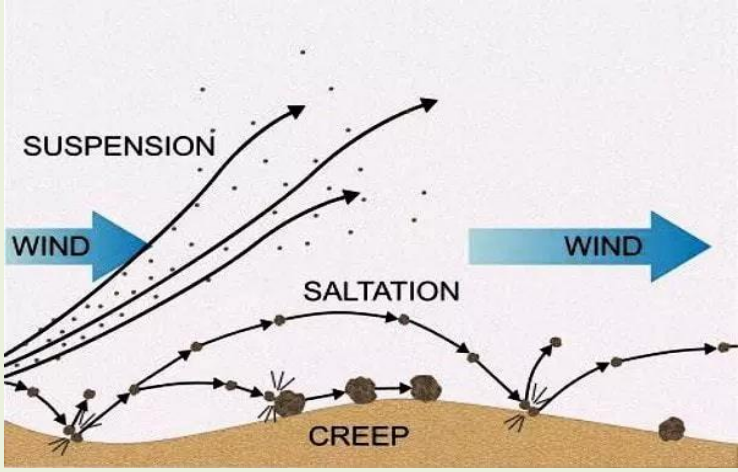
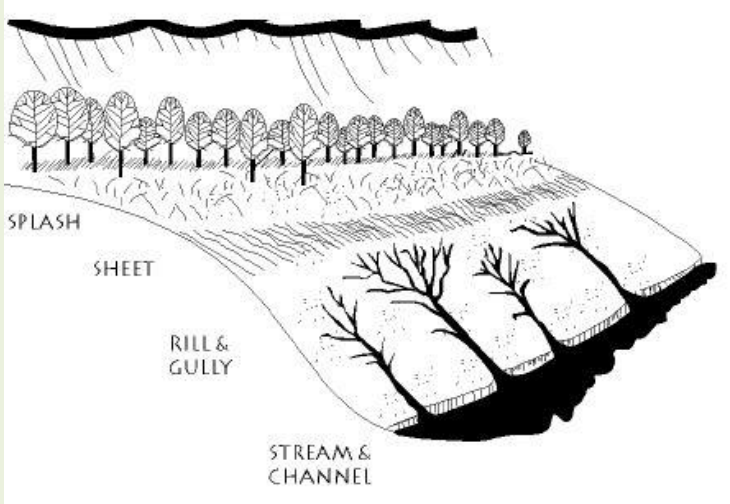
Soil Erosion

- Types
  - Normal erosion or geologic erosion
  - Accelerated or Anthropogenic erosion
- Causes
  - Climatic agents
    - Water induced erosion
      - Splash erosion
      - Sheet erosion
      - Rill erosion
      - Gully erosion
      - Slip erosion
      - Stream bank erosion
    - Wind induced erosion
      - Suspension
      - Saltation
      - Surface creep
  - Biotic agents





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Soil Erosion

- Effects
  - Decreased productivity of land
  - Desertification of land
  - Deposition of soil in water bodies
  - Reduction of agricultural land in river banks

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Soil Erosion

- Control
  - Conservational till farming
  - Stubble mulching
  - Contour farming
  - contour bunding
  - Construction of check dams
  - Terracing
  - Strip cropping
  - Alley cropping (Agro-forestry)
  - Wind breaks

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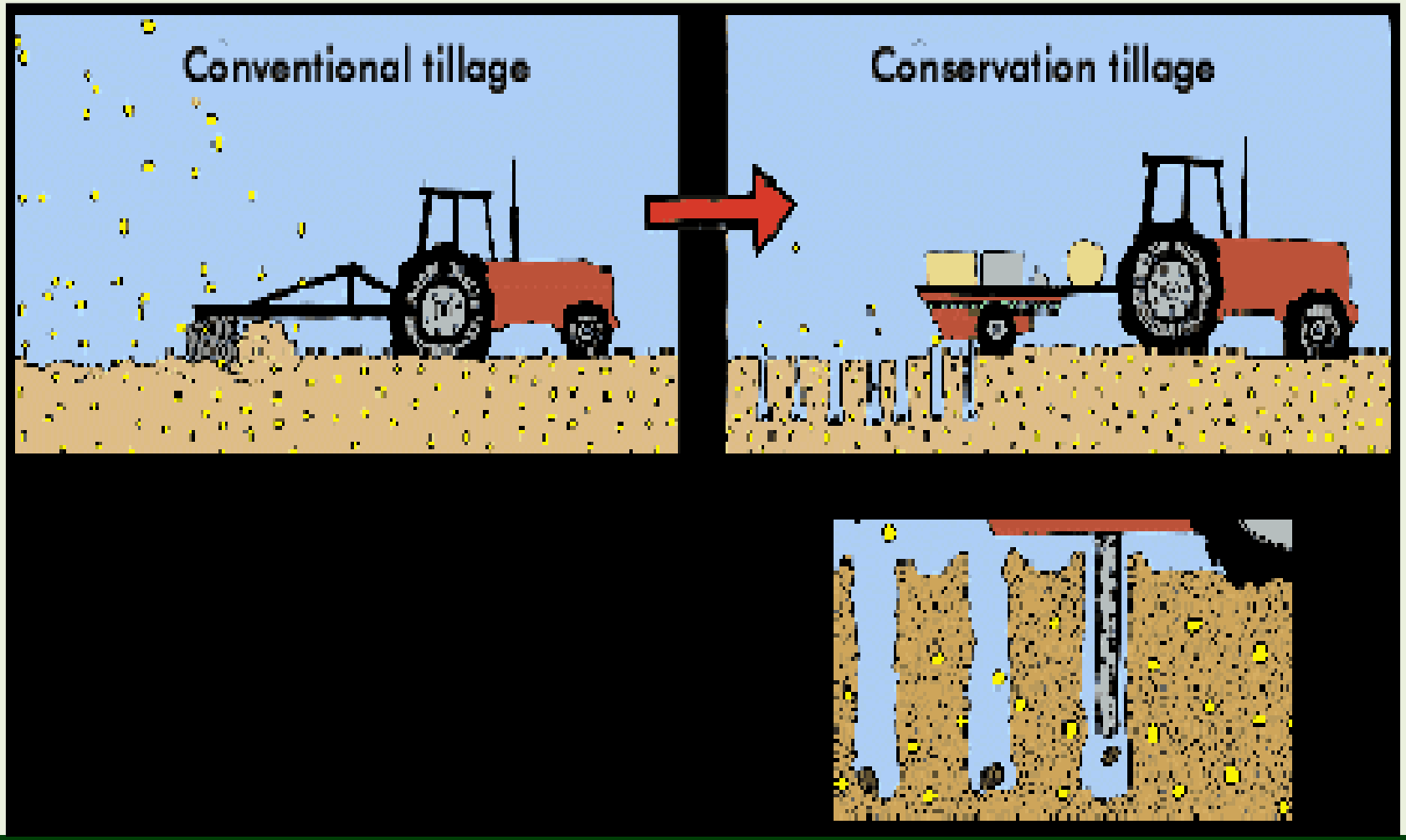
Conventional tillage VS Conservation tillage



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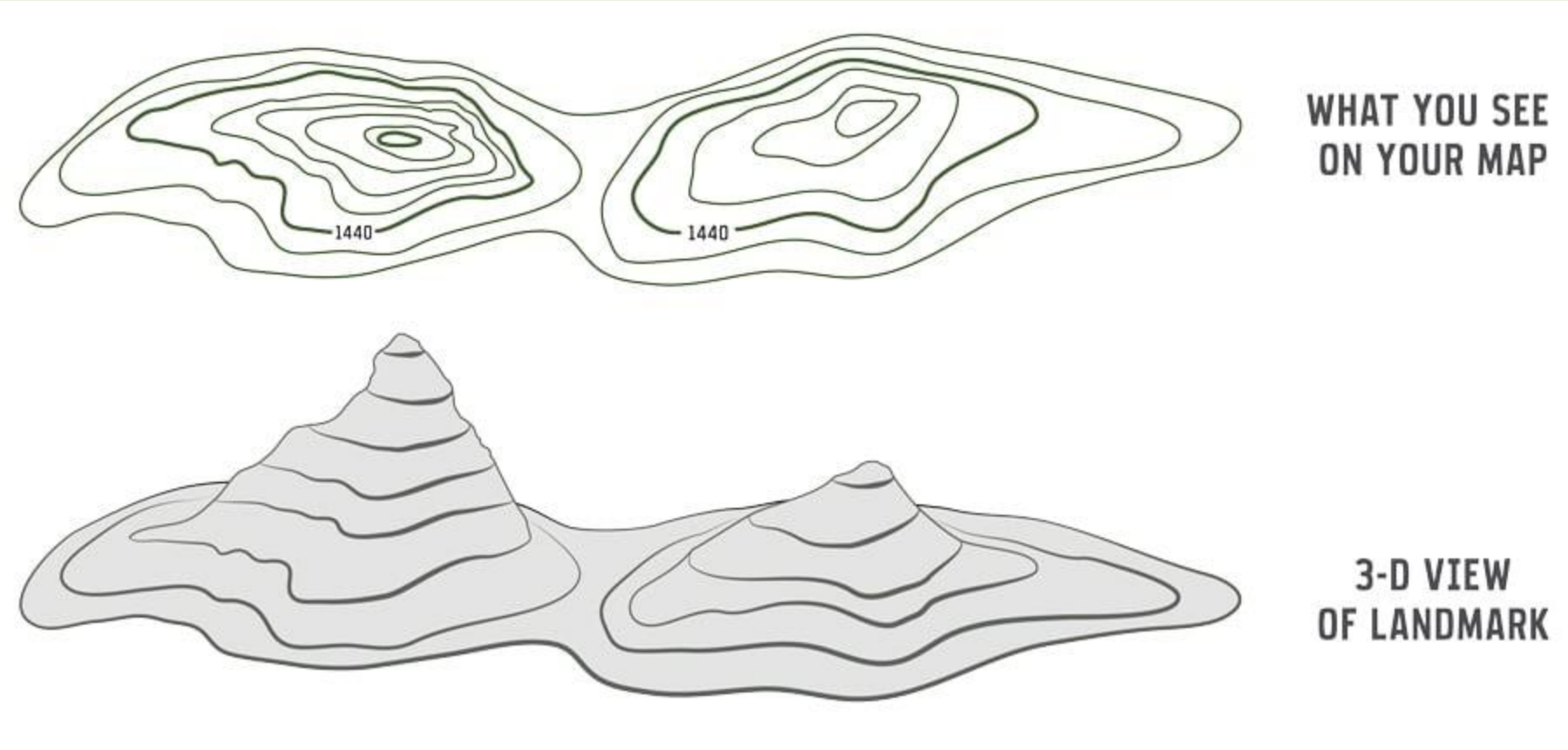
Contour Lines



Contour Lines (lines that connect the points with same altitude)

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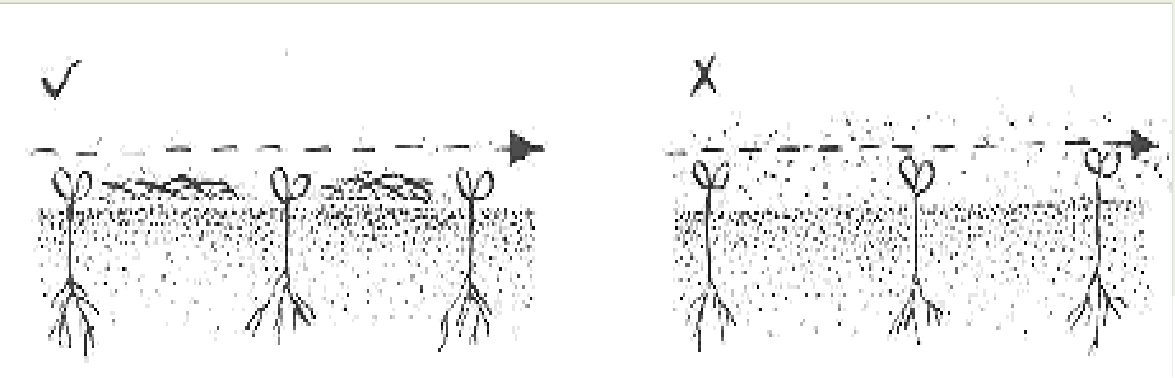
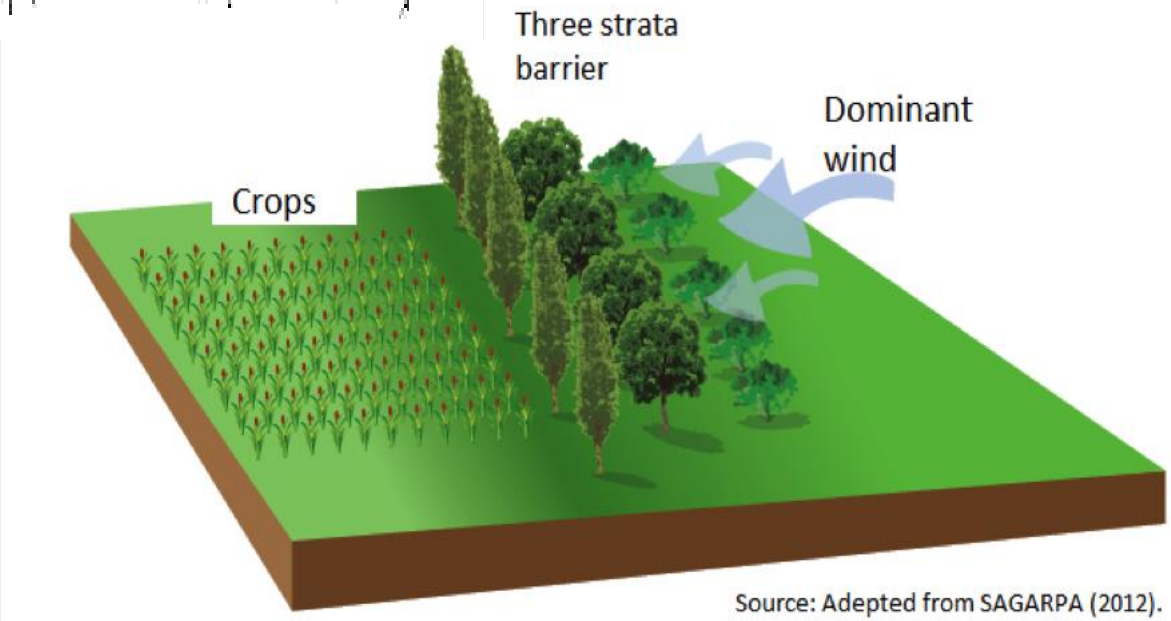
Wind Breaks





Source: Adepted from SAGARPA (2012).

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Desertification

- Types
  - Moderate (10 - 25%)
  - Severe (25 – 50%)
  - Very severe (more than 50%)
- Causes
  - Natural causes
    - Very low rain fall
    - Excessive evaporation
    - Vast difference in diurnal temperature
    - High salinity
  - Anthropogenic causes
    - Deforestation
    - Overgrazing
    - Conversion of pasture into arable land
    - Excessive use of fertilizer

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Desertification

- Effects
  - Rapid soil erosion
  - Poor soil quality
  - Unfavorable climate
  - Low water table, salty and hard water
  - Endangered human and animal life
  - Economic and human cost

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Desertification

- Control
  - Large scale plantation
  - Sustainable agricultural practices
  - Development of pasture land and controlling overgrazing
  - Development of water catchment
  - Rainwater harvesting

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Mining

- Mining is the extraction of valuable minerals or other geological materials from the Earth

Opencast Mining

Underground Mining

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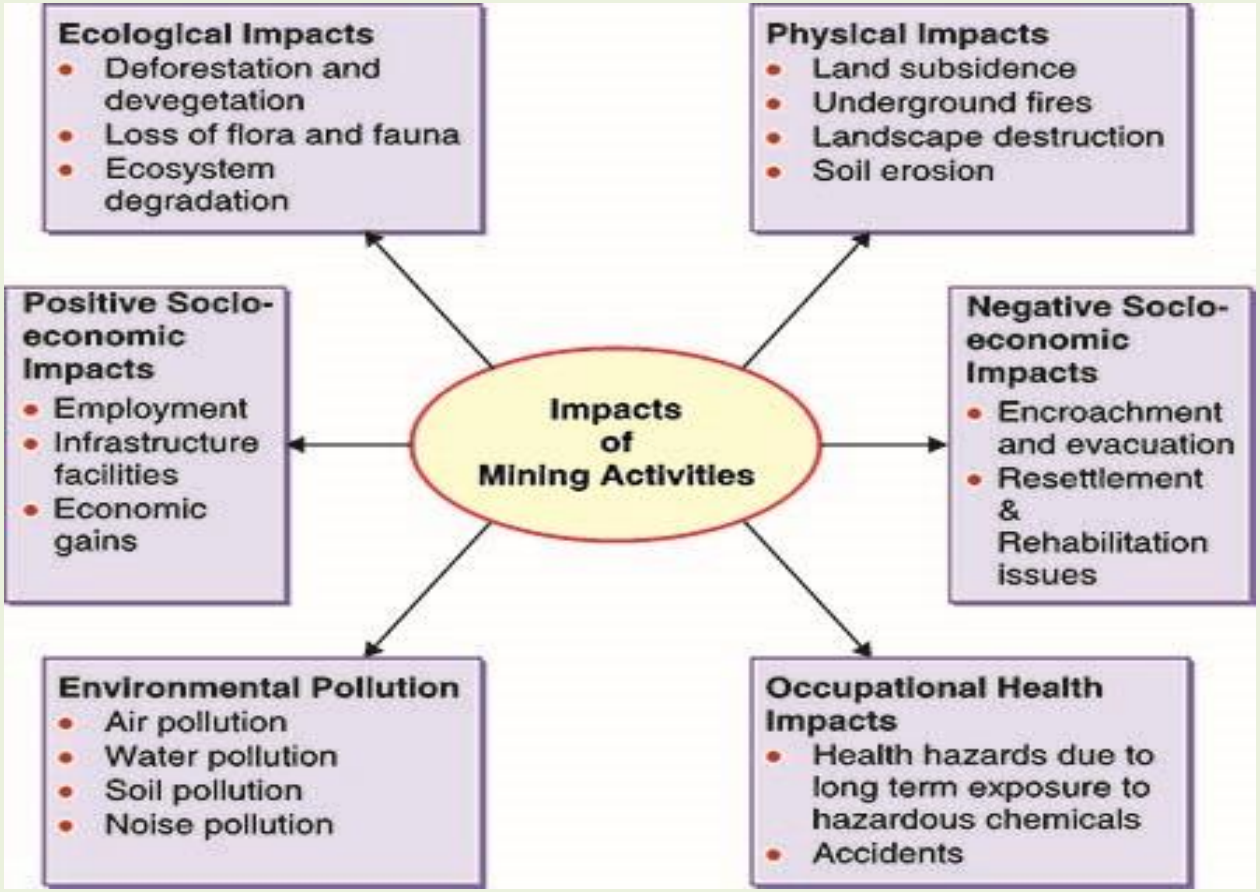
Mining

- Effects of Mining



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Thank You

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